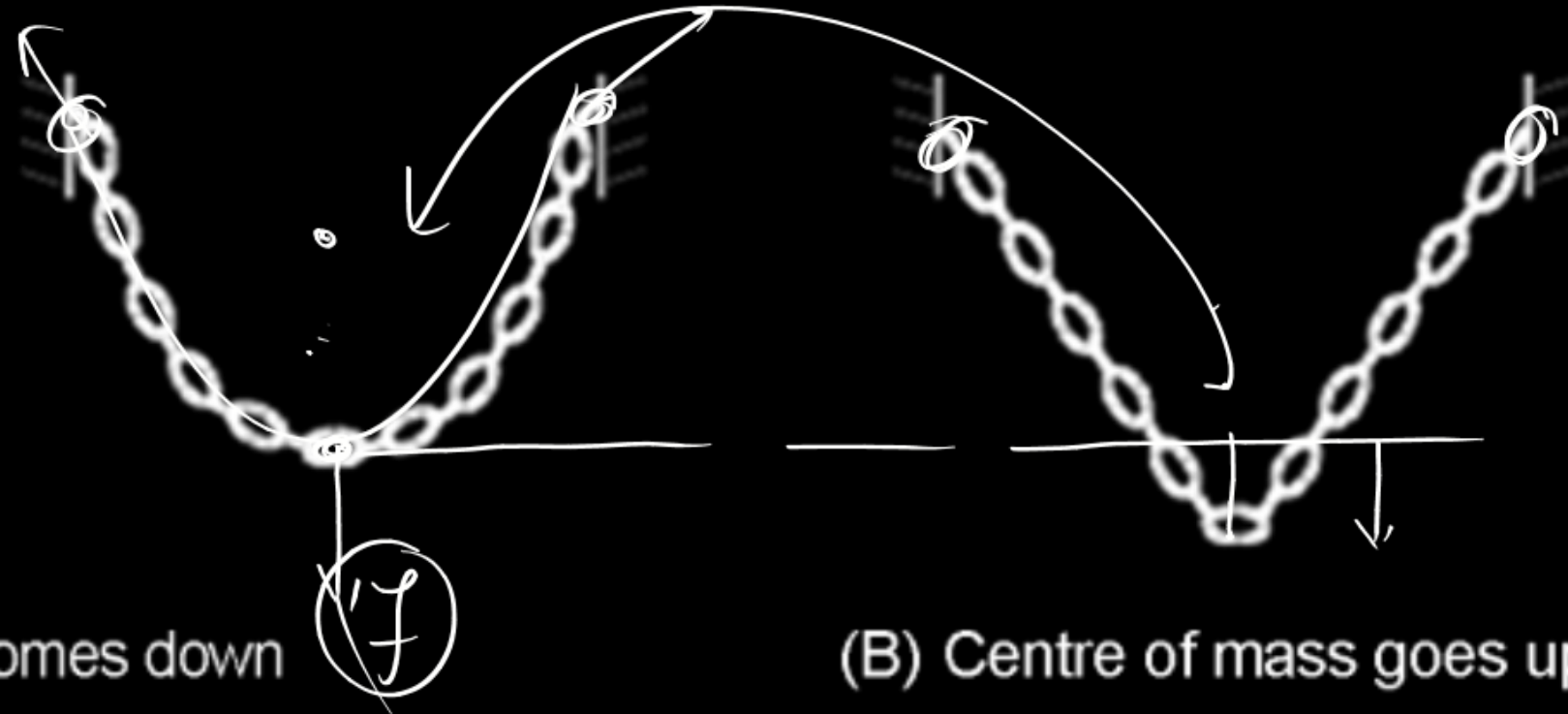


Pull down in the middle a heavy chain, fixed at both ends, in such a way that it forms a triangle. Does the centre of mass takes up a higher or lower position?



(A) Centre of mass comes down

(B) Centre of mass goes up

(C) Centre of mass remains stationary

(D) Cannot be predicted

NET

$$W_g + W_H + W_F = \Delta K$$

$$W_g + 0 + (+) = 0$$

$$W_g = \underline{(-)}$$

$$\underline{W_{ext} = +ve} = \cancel{\Delta K} + \Delta U$$

$$\underline{\Delta U = +ve}$$

$$\underline{U_f > U_i}$$

In a region, air pressure and volume are related as $P = \left(\frac{\alpha}{V^2} \right)$ where α is constant. Consider air an ideal gas of density ρ . The velocity of sound in the air is

(A) $\sqrt{\frac{P}{2\rho}}$

(B) $2\sqrt{\frac{2P}{\rho}}$

(C) $\sqrt{\frac{4P}{3\rho}}$

~~(D)~~ $V = \sqrt{\frac{2P}{\rho}}$

(9)

$$\beta = -\frac{dP}{dV} V$$

$$\beta = -\left(-\frac{2P}{V}\right) V$$

$$= \underline{\underline{2P}}$$

$$PV^2 = \alpha$$

$$P \cdot 2V dV + V^2 dP = 0$$

$$\frac{dP}{dV} = -\frac{2P}{V}$$

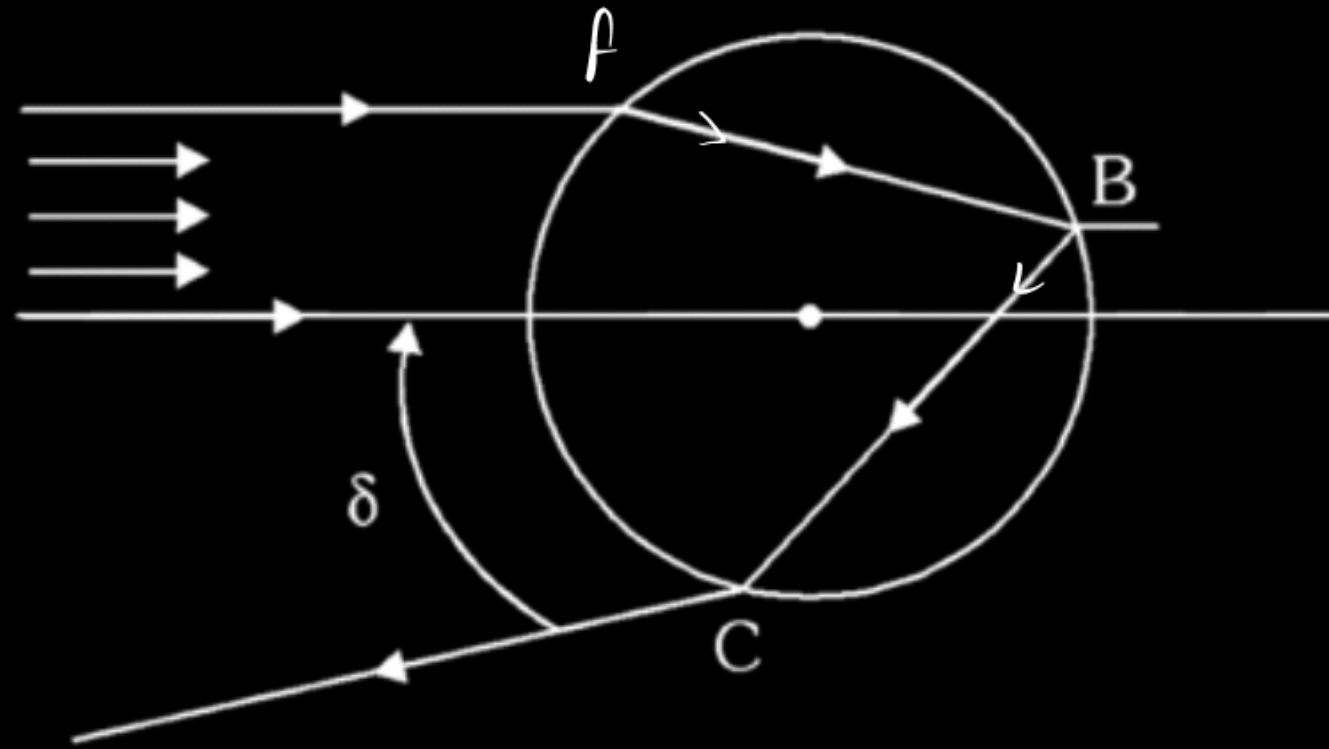
$$V = \sqrt{\frac{\beta}{\rho}} = \sqrt{\frac{2P}{\rho}}$$

$$PV^2 = \text{const}$$

$$V = \sqrt{\frac{2P}{\rho}}$$

A parallel beam of light is incident on a spherical drop of water ($\mu = 4/3$). Consider refraction of light at air-water interface, then reflection at water-air boundary (of course there will be refracted light energy in air as well), followed by refraction at water-air interface. Path of a typical ray refracted at A, then reflected at B and finally refracted at water-air interface. Path of a typical ray refracted at A, then reflected at B and finally refracted at C has been shown in the figure. Find the maximum value of angle

δ . Use: $\sin^{-1} \sqrt{\frac{20}{27}} = 60^\circ$ and $\sin^{-1} \sqrt{\frac{5}{12}} = 40^\circ$



- (A) 20° (B) 40° (C) 60° (D) 80°

$$1 \times \sin i = \frac{4}{3} \sin r$$

$$3 \cos i \, di = 4 \cos r \, dr$$

$$\frac{dr}{di} = \frac{3 \cos i}{4 \cos r} = \frac{1}{2}$$

$$\sin r = \sqrt{\frac{5}{12}}$$

$$r = 40^\circ$$

$$i = 60^\circ$$

$$3 \cos i = 2 \cos r$$

$$3 \sin i = 4 \sin r$$

$$\begin{aligned} 9 \times 1 &= 4 \cos^2 r + 16 \sin^2 r \\ &= 4 - 4 \sin^2 r + 16 \sin^2 r \end{aligned}$$

$$5 = 12 \sin^2 r$$

$$\begin{aligned} S_{\max} &= 4r - 2i \\ &= 4 \times 40 - 2 \times 60 \\ &= 160 - 120 = \underline{\underline{40^\circ}} \end{aligned}$$

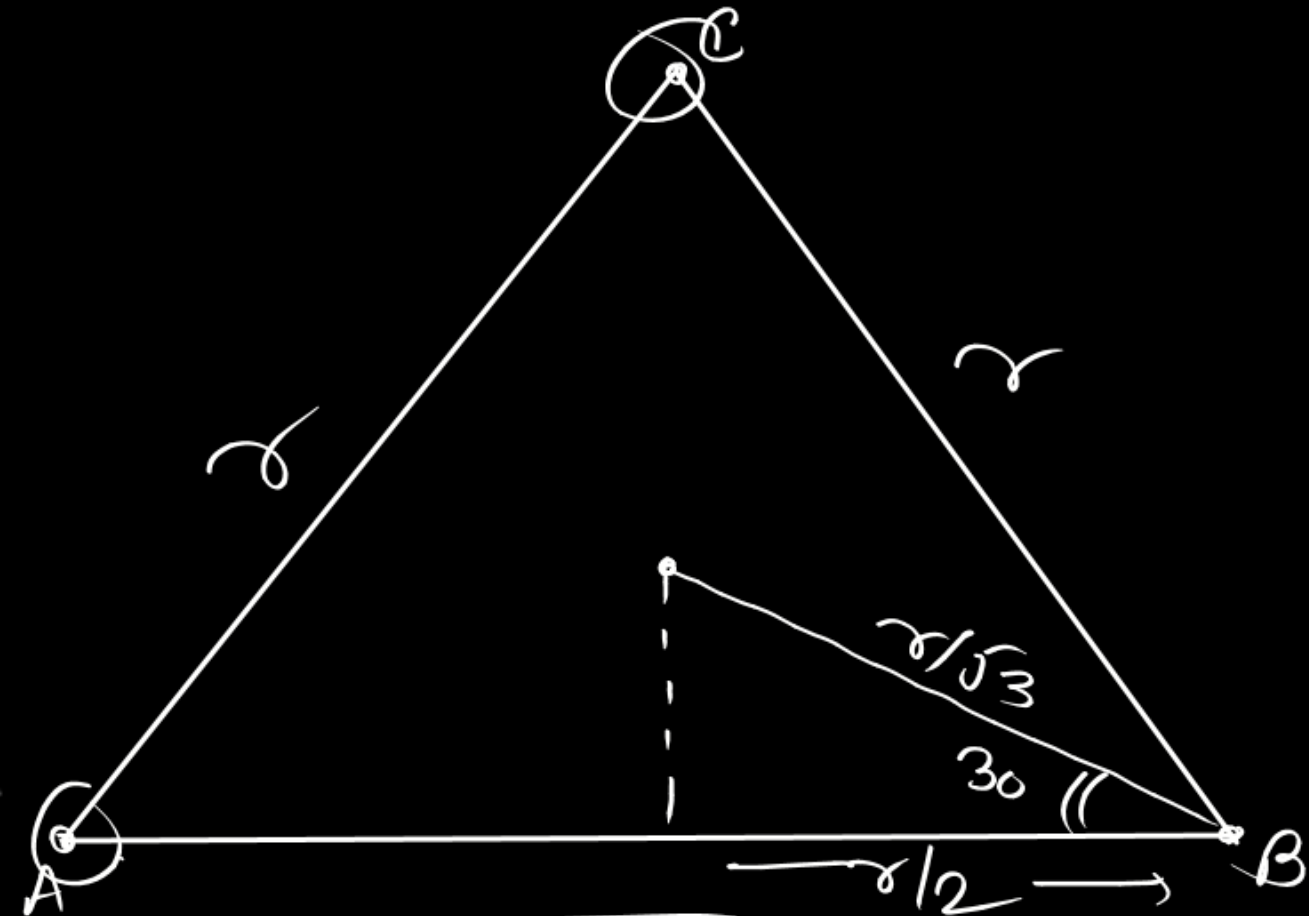
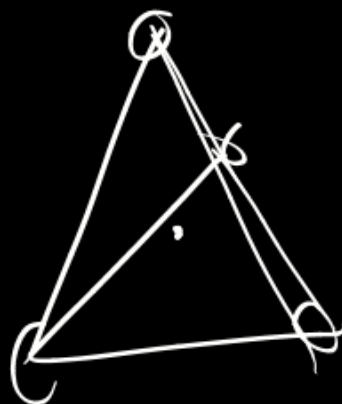
Four point masses each of mass m are placed on vertices of a regular tetrahedron. Distance between any two masses is r .

(A) Gravitation field at centre is zero

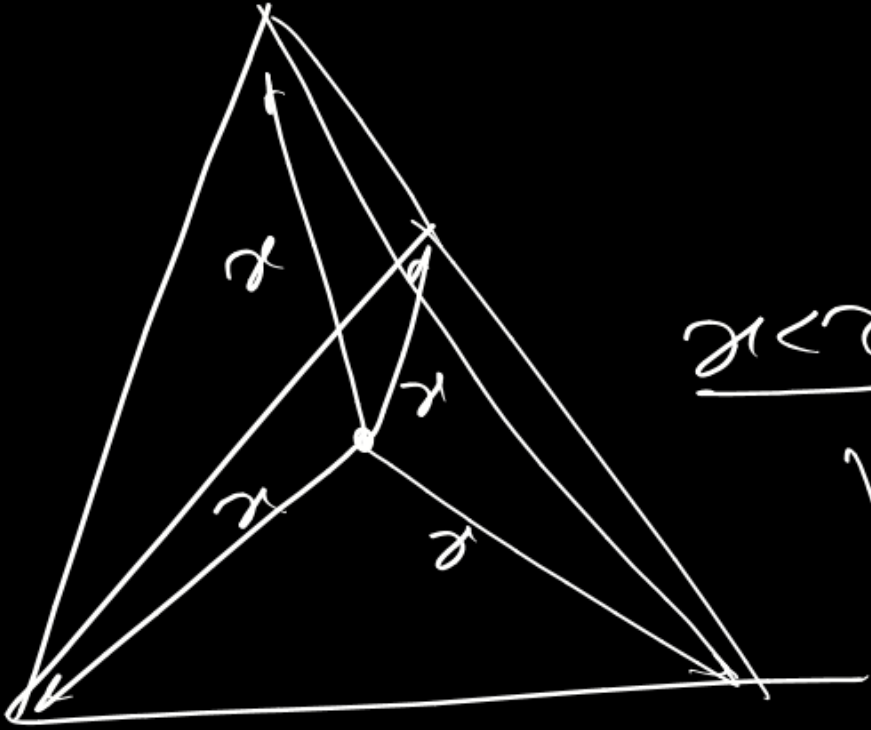
(B) Gravitation potential at centre is $\frac{-4Gm}{r}$

(C) Gravitation potential energy of system is $\frac{-6Gm^2}{r}$

(D) Gravitation force on one of the point mass is $\frac{\sqrt{6}Gm^2}{r^2}$

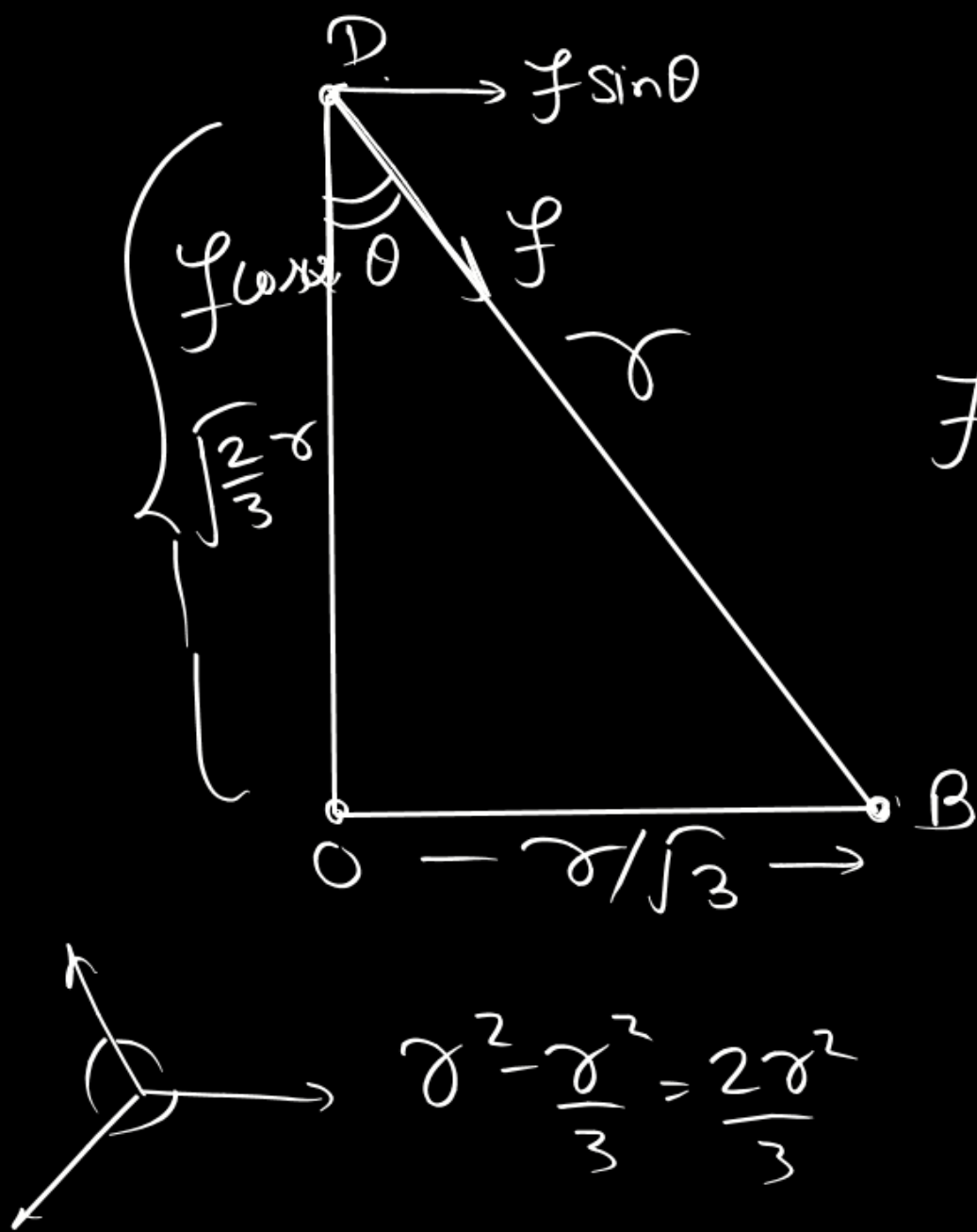
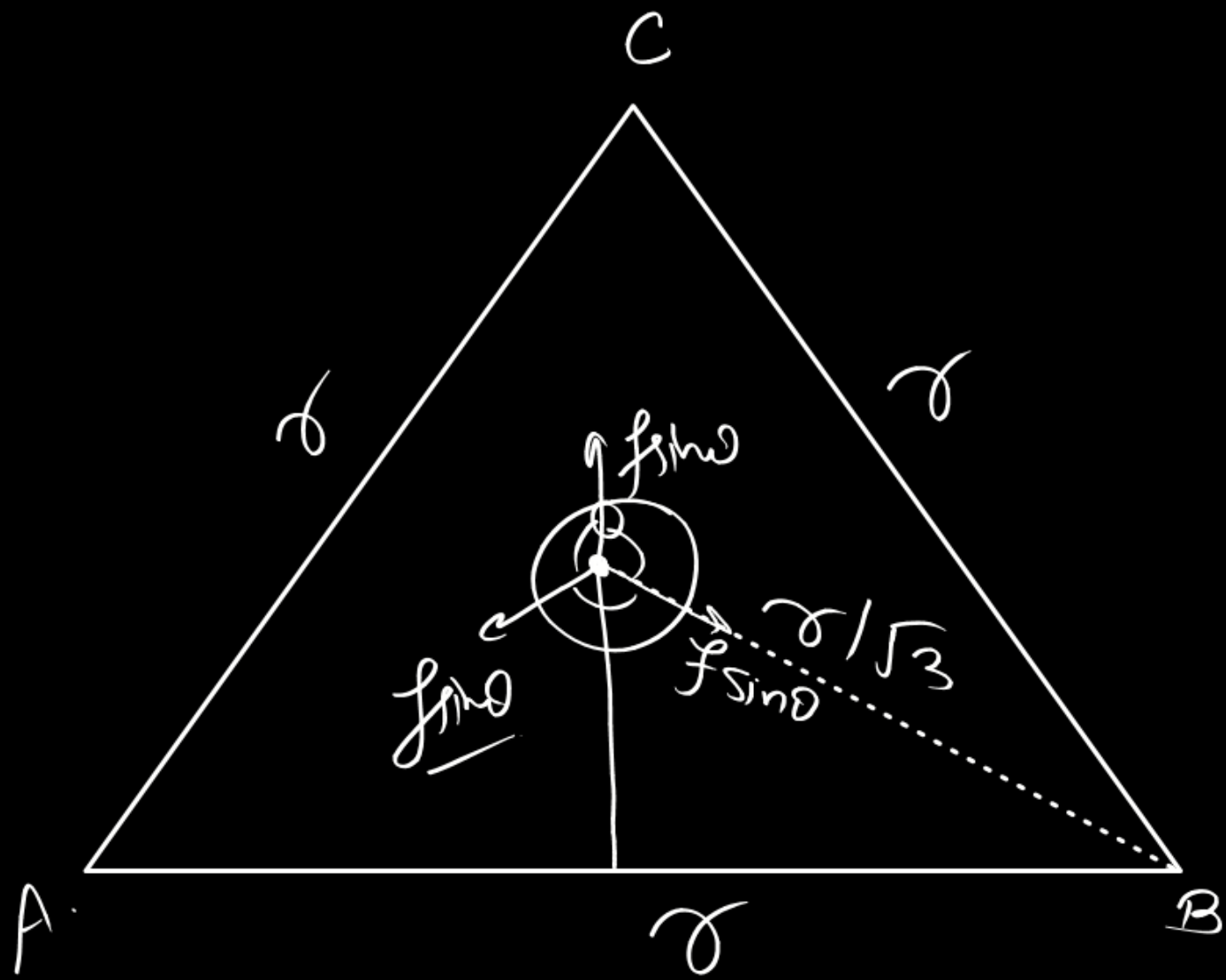


$${}^4C_2 = \frac{4!}{2!2!} = 6 \quad \boxed{U = -\frac{6m^2}{r}}$$



$$\frac{x}{r}$$

$$V = -\frac{r}{h} - \frac{r}{h} - \frac{r}{h}$$

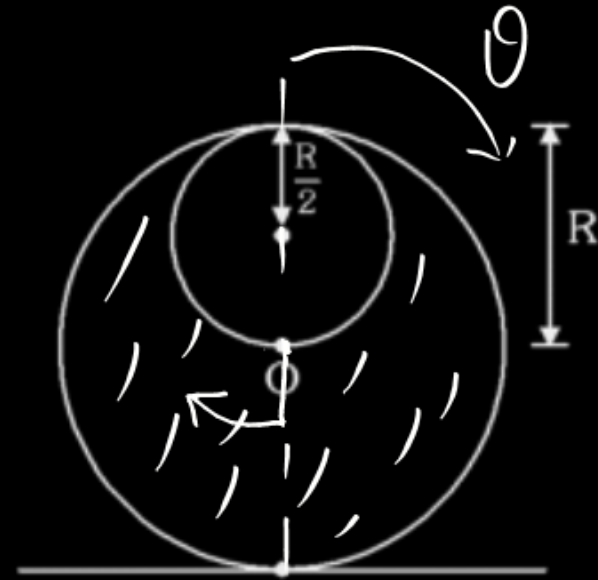


$$f = \frac{cm^2}{r^2}$$

$$\begin{aligned} F_{\text{net}} &= 3f \cos \theta \\ &= 3 \frac{cm^2}{r^2} \sqrt{\frac{2}{3}} \\ &= \frac{cm^2}{r^2} \sqrt{6} \end{aligned}$$

$$r^2 - \frac{r^2}{3} = \frac{2r^2}{3}$$

A spherical cavity of radius $\frac{R}{2}$ is removed from a solid sphere of radius R . The sphere is placed on a rough horizontal surface as shown in the figure. The sphere is given a very little push. Friction is large enough to prevent slippage.



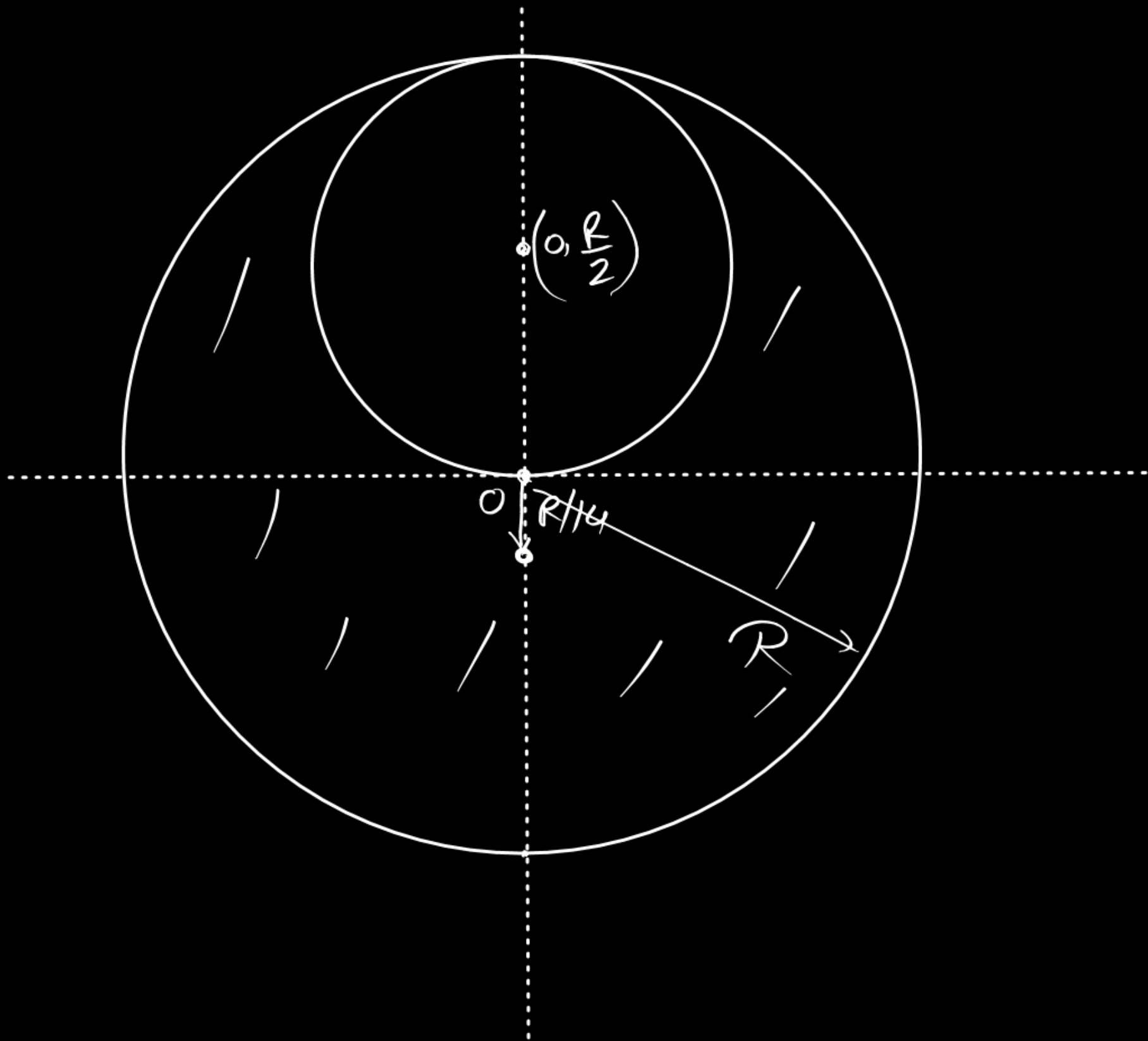
Choose the correct statement(s) :

(A) The COM of the body is at a distance of $\frac{R}{14}$ from its centre.

(B) The COM of the body is at a distance of $\frac{R}{7}$ from its centre.

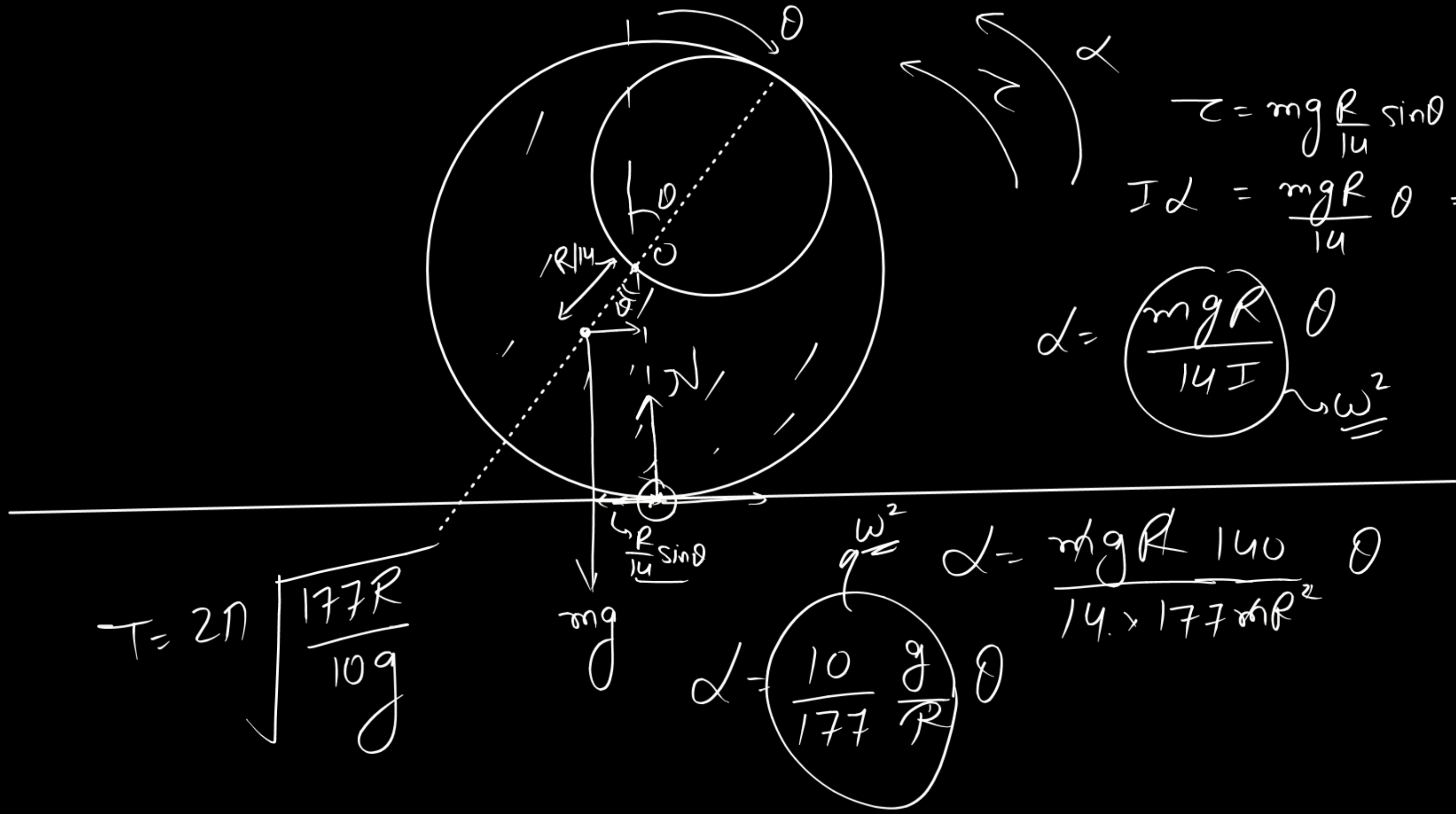
(C) The time period for very small oscillations is $T = 2\pi\sqrt{\frac{177R}{10g}}$

(D) The time period for very small oscillations is $T = 2\pi\sqrt{\frac{32R}{5g}}$



$$x_{com} = \frac{V \times 0 - \frac{V}{8} \times 0}{V - \frac{V}{8}} = 0$$

$$y_{com} = \frac{V \times 0 - \frac{V}{8} \times \frac{R}{2}}{\cancel{V - \frac{V}{8}} \frac{7V}{8}} = -\frac{R}{14}$$



$$\tau = mg \frac{R}{14} \sin \theta$$

$$I \alpha = \frac{mgR}{14} \theta \Rightarrow \underline{\underline{\text{sum}}}$$

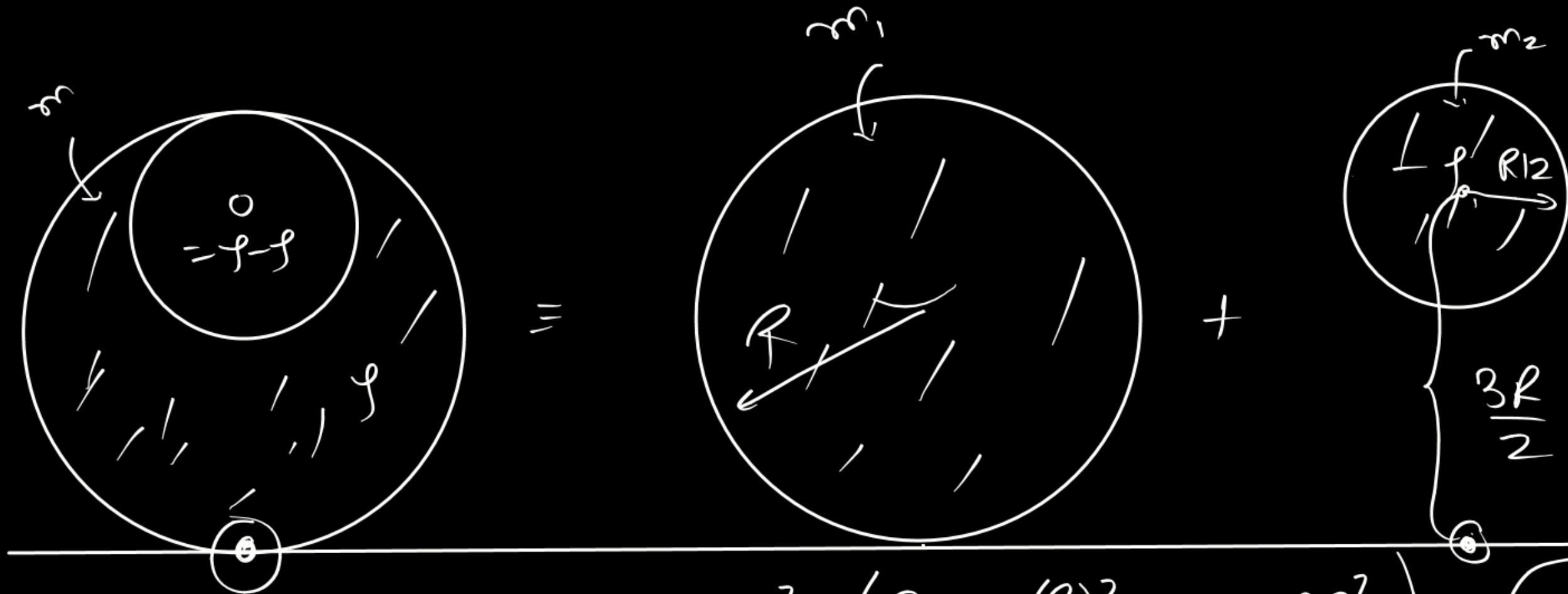
$$\alpha = \left(\frac{mgR}{14I} \right) \theta \sim \omega^2$$

$$T = 2\pi \sqrt{\frac{177R}{10g}}$$

$$\alpha = \left(\frac{10}{177} \frac{g}{R} \right) \theta$$

$$\alpha = \frac{mgR}{14 \times 177 R^2} \theta$$

$$\rho = \frac{\frac{33}{4}\pi \frac{7R^3}{82}}{7\pi R^3} = \frac{6m}{7\pi R^3}$$



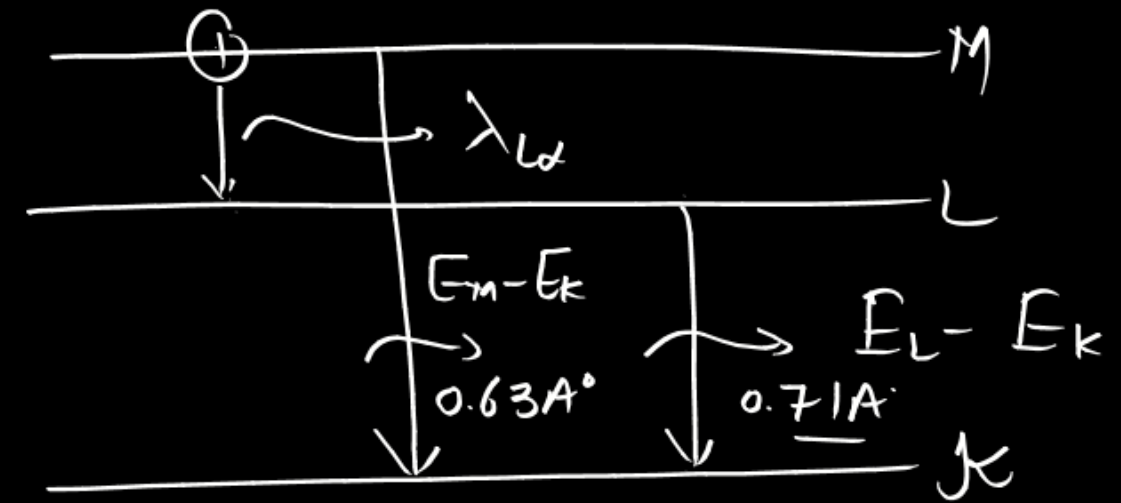
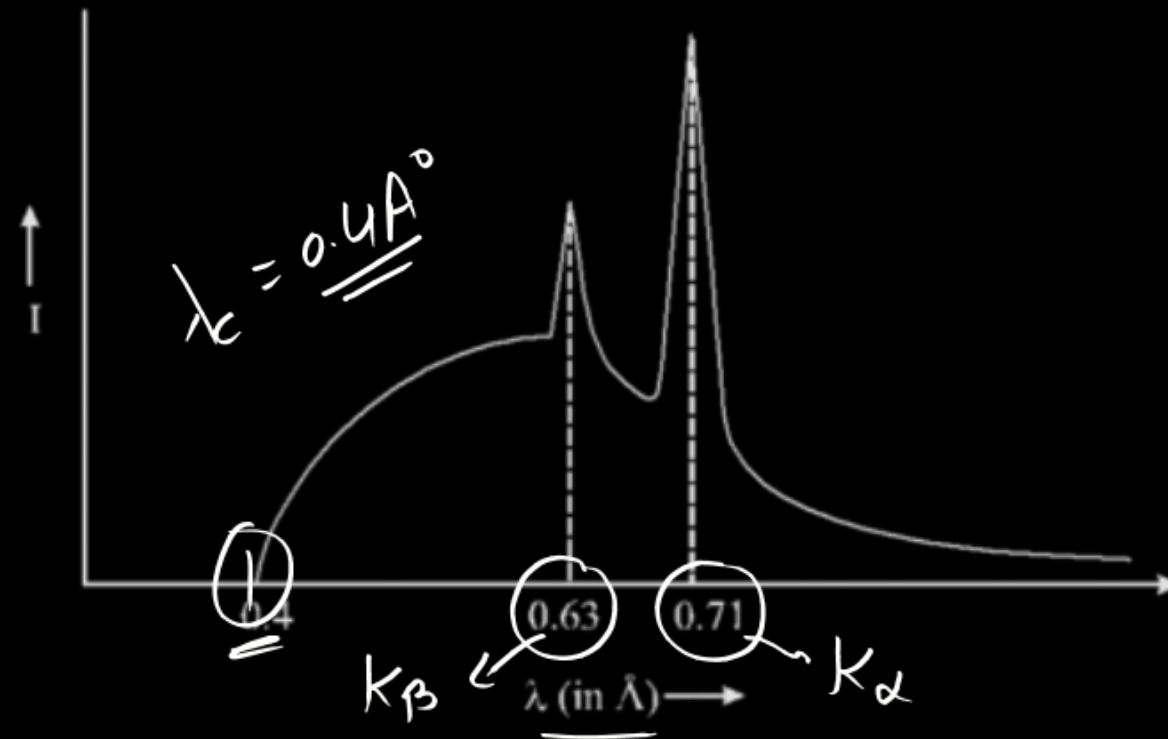
$$I = I_f + I_{-f} = \frac{7}{5} m_1 R^2 + \left(\frac{2}{5} m_2 \left(\frac{R}{2} \right)^2 + m_2 \frac{9R^2}{4} \right)$$

$\downarrow$ $\frac{4}{3}\pi R^3$ $\downarrow$ $\frac{4}{3}\pi \left(\frac{R}{2} \right)^3$

$$I = \frac{177}{140} m R^2$$

Figure shows intensity versus wavelength graph of X-rays coming from Coolidge-tube with molybdenum as anticathode:

$$\lambda = \frac{0.63 \times 0.71}{0.08} = 5.60 \text{ \AA}$$



$$E_M - E_K = \frac{hc}{0.63 \text{ \AA}}$$

$$E_L - E_K = \frac{hc}{0.71 \text{ \AA}}$$

$$E_M - E_L = \frac{hc}{\lambda} = \frac{hc}{0.63 \text{ \AA}} - \frac{hc}{0.71 \text{ \AA}}$$

The two peaks shown in graph correspond to K_α and K_β X-rays

- ☒ (A) Wavelength of L_α X-rays from Coolidge tube will be (approximately) $5.60 \text{ \AA}$
- ☐ (B) Voltage applied across Coolidge tube in (approximately) 30 kV
- ☒ (C) Voltage applied across Coolidge tube in (approximately) 31 kV
- ☒ (D) Wavelength of L_α X-rays from Coolidge tube will be (approximately) $4.26 \text{ \AA}$

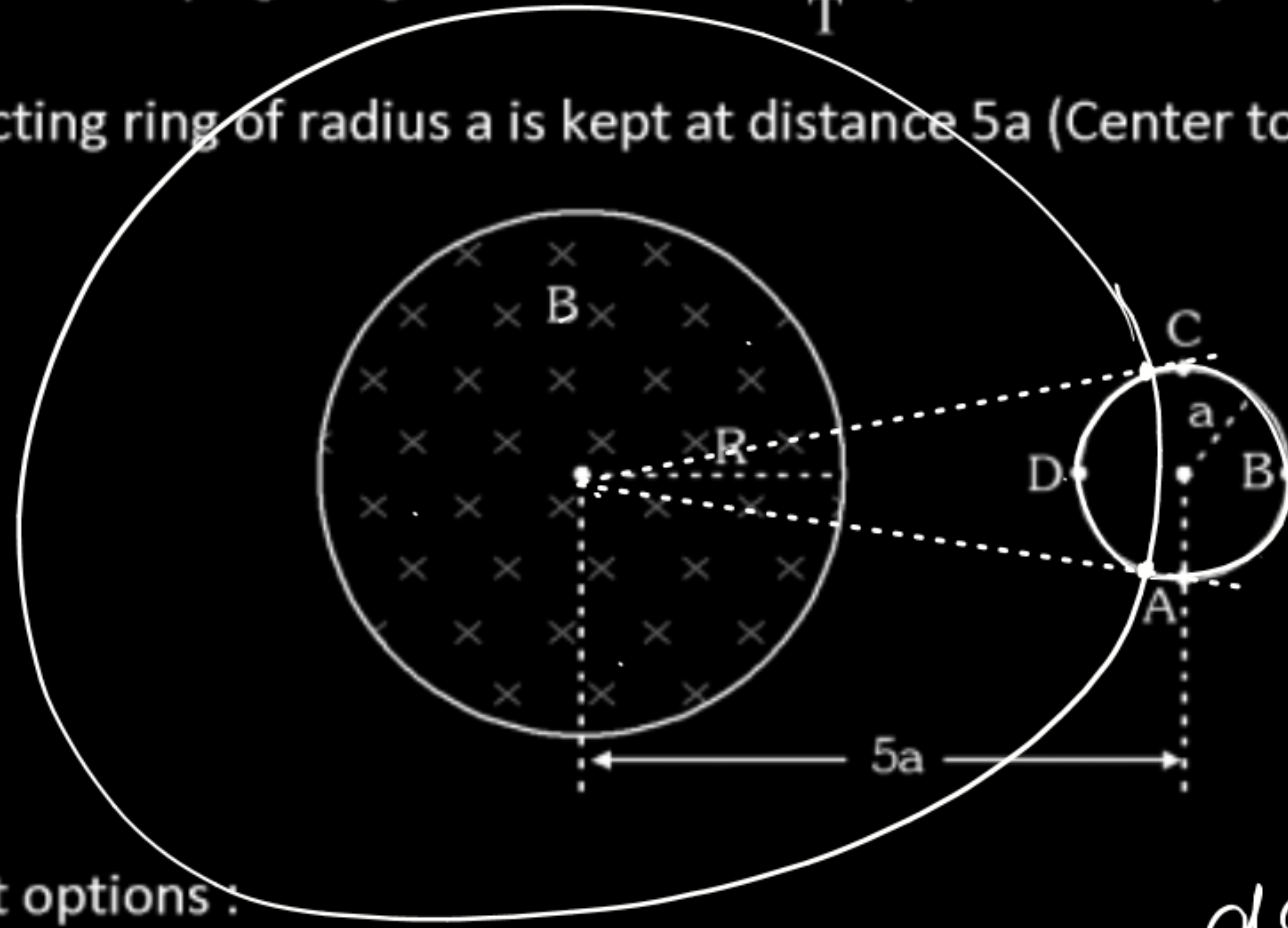
$$V_a e = \frac{hc}{\lambda_c}$$

$$\lambda_c = \frac{hc}{e V_a} = \frac{12420 \text{ V-Å}^{\circ}}{V_a}$$

$$V_a = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 0.4 \times 10^{-10}}$$

$$V_a = \frac{12420}{\lambda_c (\text{Å}^{\circ})}$$

Uniform but time varying magnetic field $B = \frac{B_0 t}{T}$ present in cylindrical region of radius R . There is a circular conducting ring of radius a is kept at distance $5a$ (Center to center distance; $T = \text{constant}$).



Choose correct options :

~~(A)~~ The value of emf in ring from B to C is zero. ~~X~~

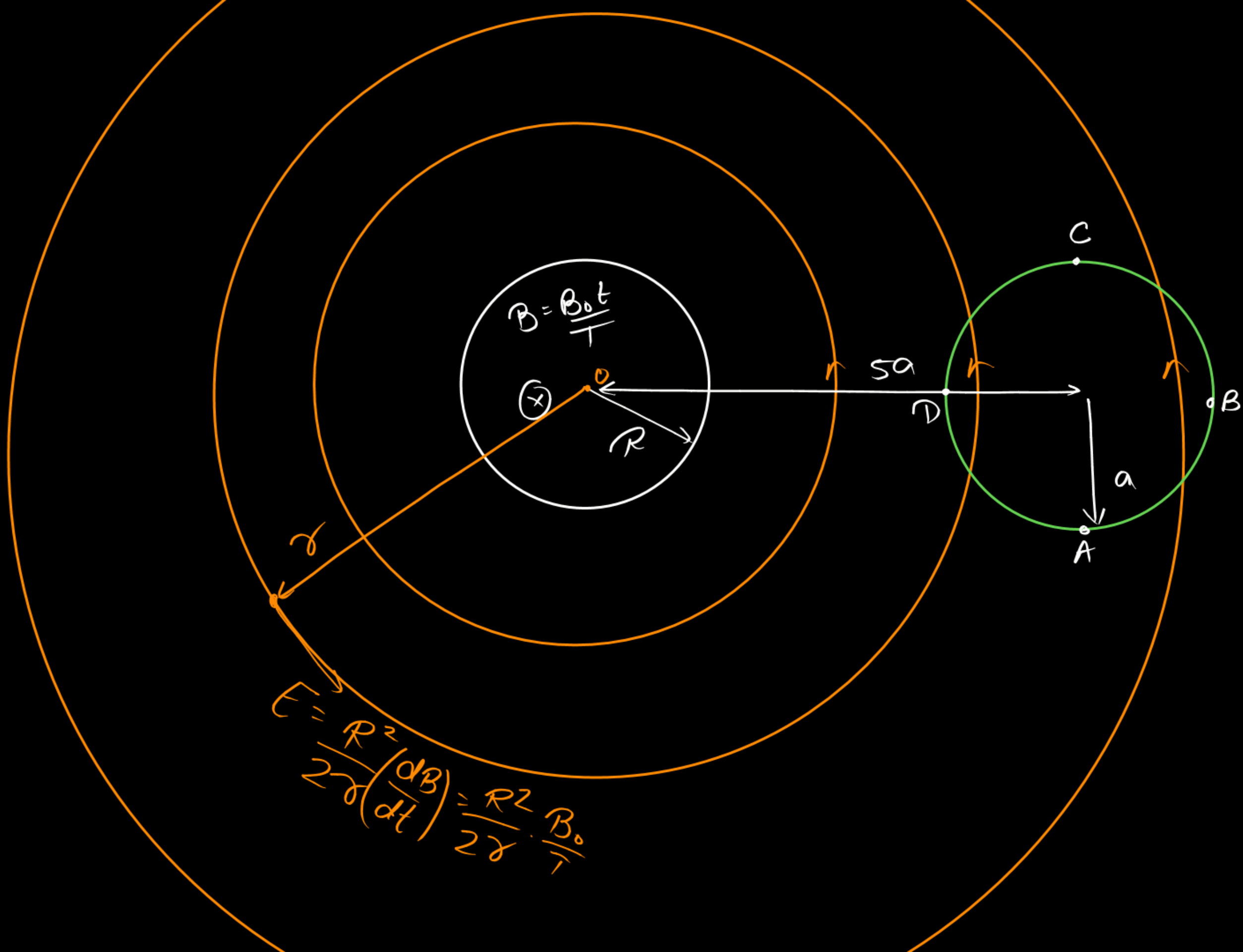
(B) The value of emf in ring from B to D is zero.

(C) The maximum value of emf in ring from one point on the ring to other point on the ring is

$$\frac{R^2 B_0}{T} \sin^{-1} \left(\frac{1}{5} \right)$$

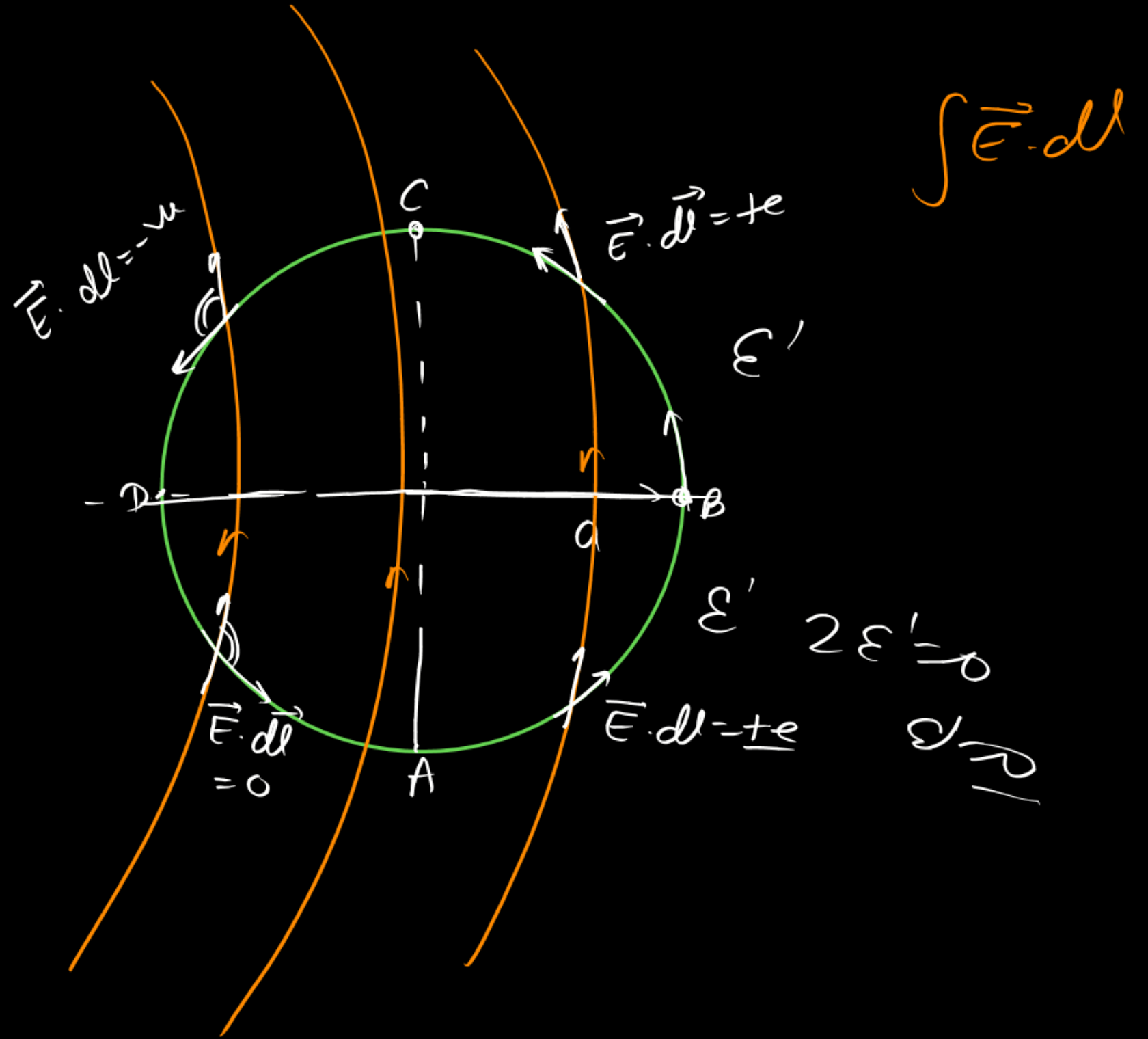
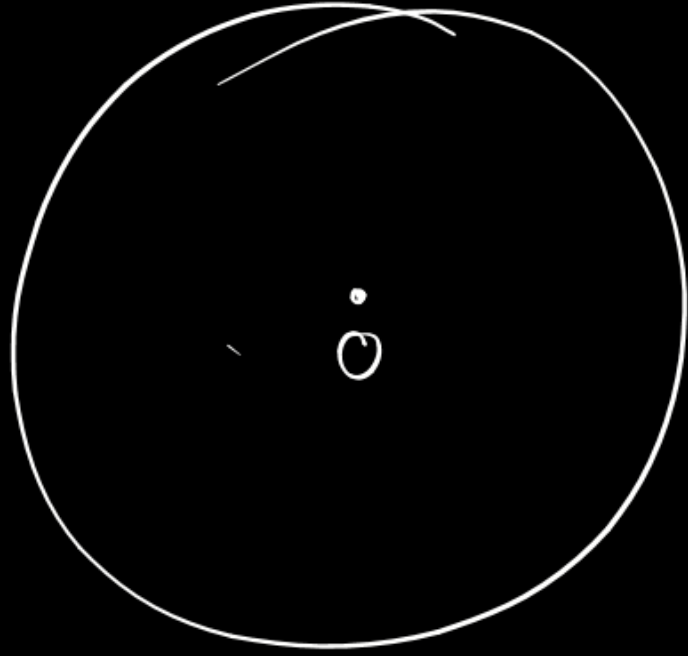
(D) The value of emf in ring from A to C is zero.

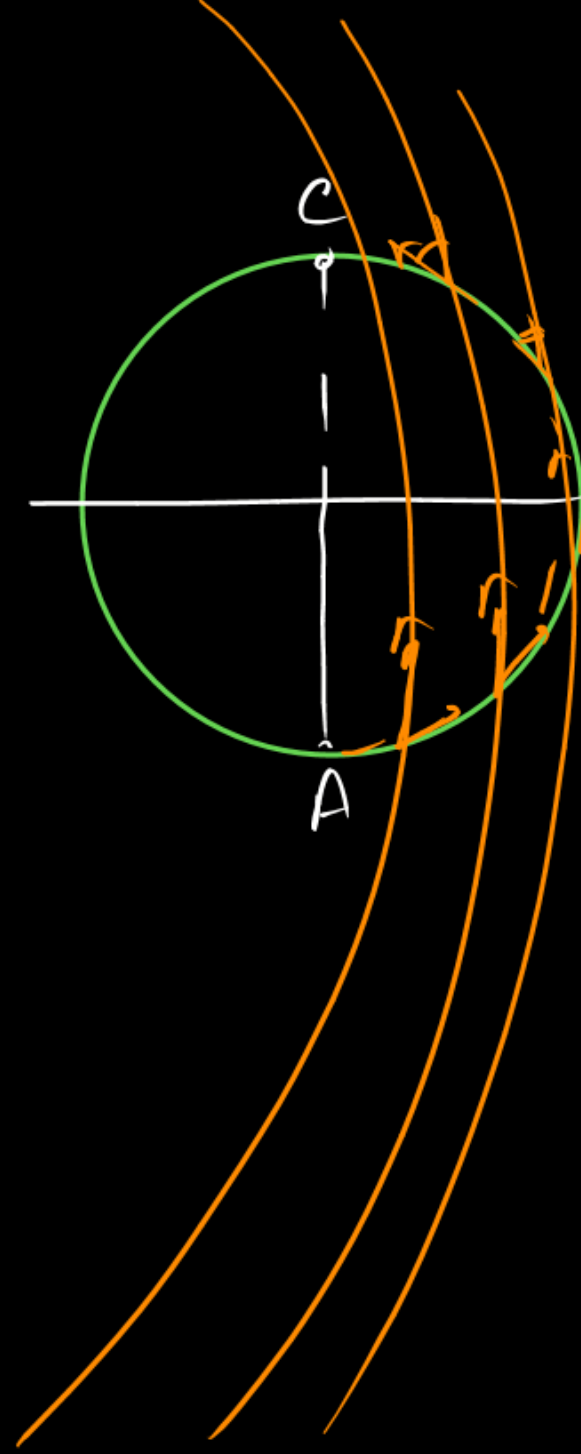
$$d\mathcal{E} = dV = \underline{\vec{E} \cdot d\vec{l}}$$

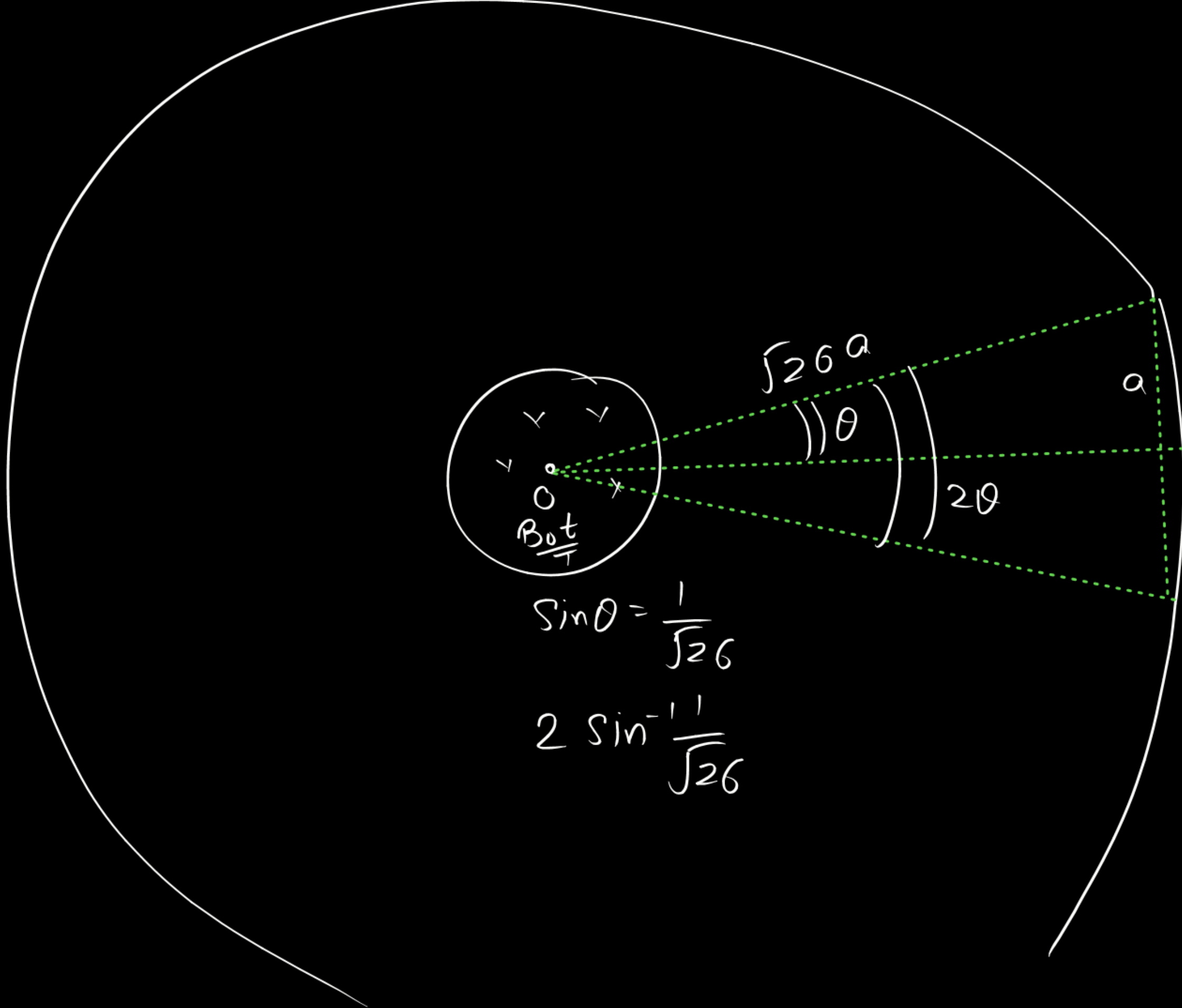


$$F = \frac{R^2}{2\gamma} \left(\frac{dB}{dt} \right) = \frac{R^2}{2\gamma} \cdot \frac{B_0}{T}$$

$$\oint_{ABCD} \vec{E} \cdot d\vec{l} = 0$$







$$\sin \theta = \frac{1}{\sqrt{26}}$$

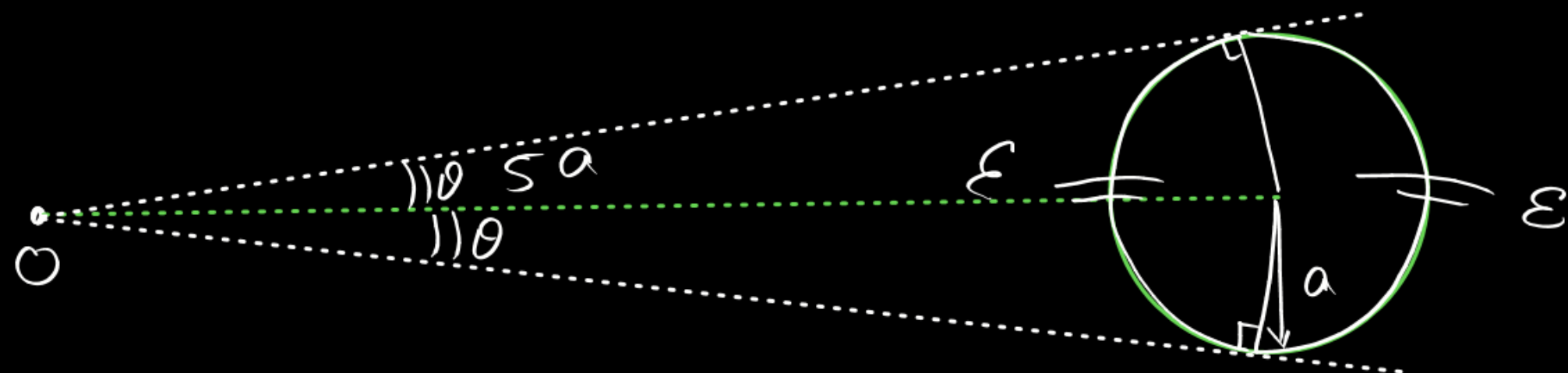
$$2 \sin^{-1} \frac{1}{\sqrt{26}}$$

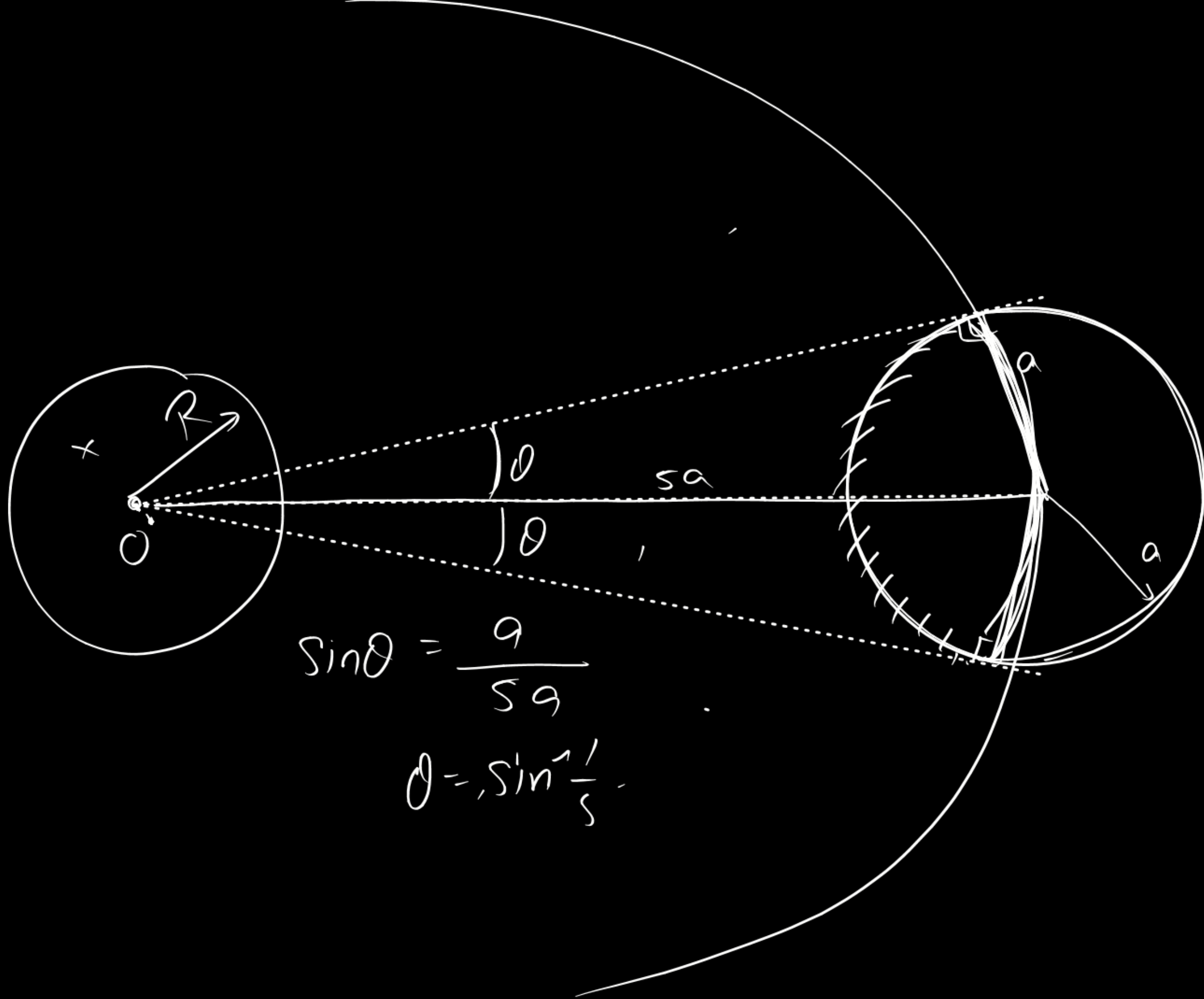
$$\phi = \frac{B_0 t}{T} \pi R^2$$

$$\mathcal{E} = \frac{B_0}{T} \pi R^2$$

$$2\pi \rightarrow \frac{B_0}{T} \pi R^2$$

$$\mathcal{E}' = \frac{\frac{B_0}{T} \pi R^2}{2\pi} 2 \sin^{-1} \frac{1}{\sqrt{26}}$$





$$\sin \theta = \frac{a}{5a}$$

$$\theta = \sin^{-1} \frac{1}{5}$$

$$\phi = \frac{B_0 t}{T} \pi R^2$$

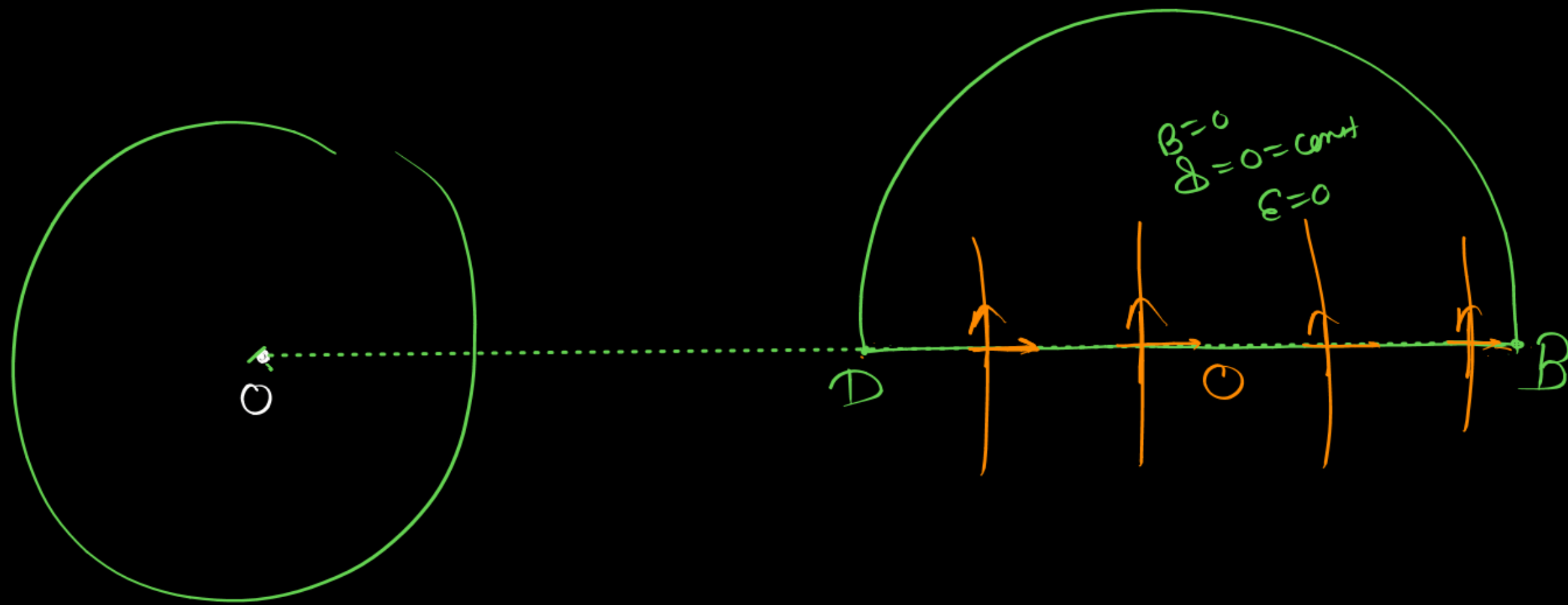
$$\epsilon = \frac{B_0}{T} \pi R^2$$

$$2\pi \rightarrow \frac{B_0}{T} \pi R^2$$

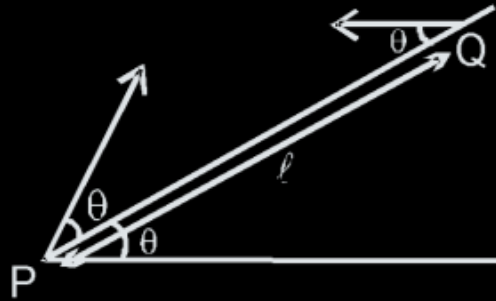
$$2\theta \rightarrow \frac{\frac{B_0}{T} \pi R^2}{2\pi} \times 2\theta$$

$$\frac{B_0 R^2 \theta}{T}$$

$$\frac{B_0 R^2}{T} \sin^{-1} \left(\frac{1}{5} \right)$$



From an incline plane two particles P and Q are projected with same speed at same angle θ , one up and other down the incline plane as shown in figure. Which of the following statement(s) is/are correct ? (Assume both particle moves in same plane)



(A) The particles will collide the plane with same speed

(B) The particles will collide in mid-air before reaching the incline plane, if projected simultaneously

and $l < \frac{4u^2 \sin \theta}{g}$

(C) Both particles strike the plane normally

(D) The time of flight of each particle is same

A boat moves relative to river with a velocity which is n times the river flow velocity.

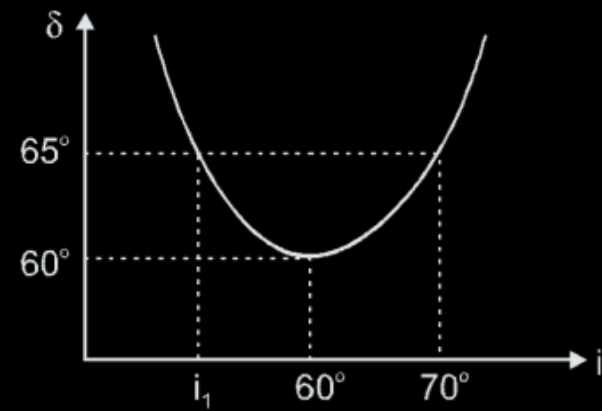
(A) If $n < 1$, boat cannot cross the river

(B) If $n = 1$, boat cannot cross the river without drifting

(C) If $n > 1$, boat can cross the river along shortest path

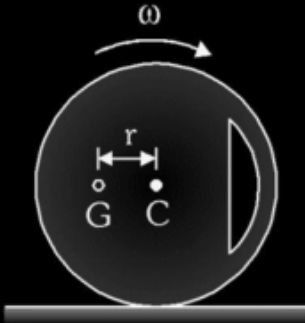
(D) Boat can cross the river whatever is the value of n (excluding zero)

The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Pick up the correct statements.



- (A) The angle of prism is 60°
- (B) The refractive index of the prism is $n = \sqrt{3}$
- (C) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
- (D) The curve of ' δ ' vs ' i ' is parabolic

The mass center G of a 3 kg wheel of radius $R = 180\text{ mm}$ is located at a distance $r = 60\text{ mm}$ from its geometric center C. The centroidal radius of gyration of the wheel is $k = 90\text{ mm}$. As the wheel rolls without sliding, its angular velocity is observed to vary. Angular speed of wheel is $\omega = 8\text{ rad/s}$ in the position shown

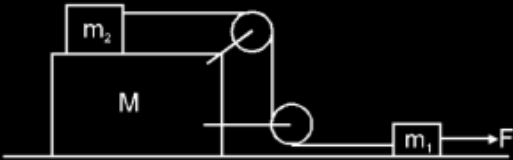


The angular velocity of the wheel when the mass center G is directly above the geometric center C is close to :

- (A) 20 rad/s. (B) 15 rad/s. (C) 10 rad/s. (D) 5 rad/s.

Match the following :

Three blocks of masses m_1 , m_2 and M are arranged as shown in figure. All the surfaces are frictionless and string is inextensible. Pulleys are light. A constant force F is applied on block of mass m_1 . Pulleys and string are light. Part of the string connecting both pulleys is vertical and part of the strings connecting pulleys with masses m_1 and m_2 are horizontal.

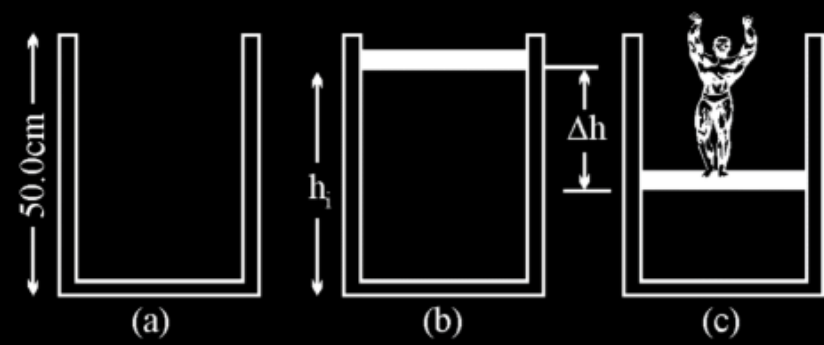


List-I	List-II
(P) Acceleration of mass m_1	(1) $\frac{F}{m_1}$
(Q) Acceleration of mass m_2	(2) $\frac{F}{m_1 + m_2}$
(R) Acceleration of mass M	(3) zero
(S) Tension in the string	(4) $\frac{m_2 F}{m_1 + m_2}$

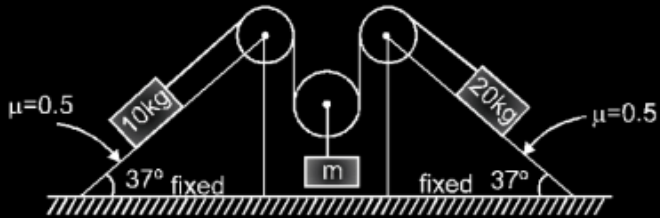
	P	Q	R	S
(A)	2	2	3	4
(B)	2	1	3	4
(C)	2	2	4	1
(D)	2	1	3	1

Two persons A and B wear glasses of optical powers (in air) $P_1 = + 2 \text{ D}$ and $P_2 = + 1 \text{ D}$ respectively. The glasses have refractive index 1.5. Now they jump into a swimming pool and look at each other. B appears to be present at distance 2m (from A) to A. A appears to be present at distance 1m (from B) to B. The refractive index of water in the swimming pool, in the form $\frac{X}{5}$ and find X.

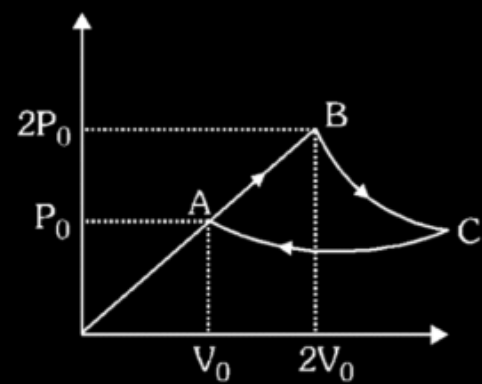
A highly conducting cylinder that has a cross-section area of 100cm^2 and is 50cm deep is filled with air at 21.0°C and 1.00 atm (figure-a) A 20.0 kg piston is now lowered into the cylinder, compressing the air trapped inside (figure-b). Finally, a 80.0 kg man stands on the piston, further compressing the air, which remains at 21°C (figure-c). What temperature T (in $^\circ\text{C}$) should the gas be maintained to raise the piston and the man back to h_1 ? ($g = 10\text{ m/s}^2$). Fill T/31 in OMR



In given arrangement, 10 kg and 20 kg blocks are kept at rest on two fixed inclined planes. All strings and pulleys are ideal. The difference of maximum and minimum values of m for which system remain in equilibrium is: ($g = 10 \text{ m/s}^2$)

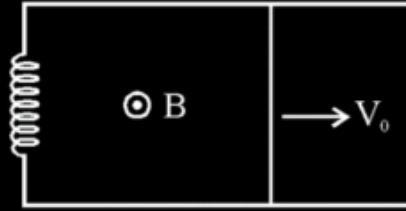


One mole of a diatomic gas is taken around a cyclic process shown in the figure. The process BC is adiabatic expansion and the process CA is isothermal compression. If the volume of the gas at C is given by $V_C = (2)^n V_0$, then the value of n is

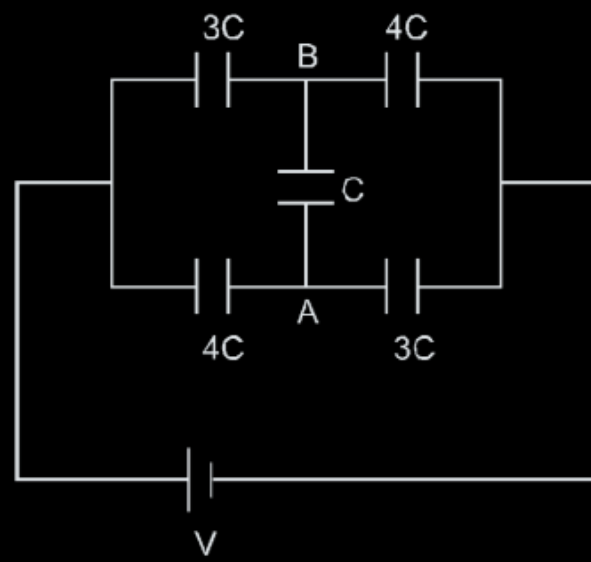


A particle is thrown with initial speed $u = 1\text{m/s}$ in vertically upward direction. Due to air resistance, retardation produced in the particle is proportional to square of velocity of particle i.e. αv^2 , where α is 10 m^{-1} . If time taken by particle to reach maximum height is $\frac{\pi}{x}$ sec then, x is (Take $g = 10\text{m/s}^2$)

A loop is formed by two parallel conductors connected by a solenoid with inductance $L = 10^3 \text{ H}$ and a conducting rod of mass $m = 1 \text{ gm}$ which can freely slide (without friction) over a pair of conductors. The conductors are located in a horizontal plane in a uniform magnetic field $B = 4\text{T}$ in the direction shown. The distance between conductors is equal to $\ell = 1 \text{ m}$. At the moment $t = 0$, the rod is imparted an initial velocity V_0 to the right. Determine angular frequency (in rad/s) of oscillation of rod.

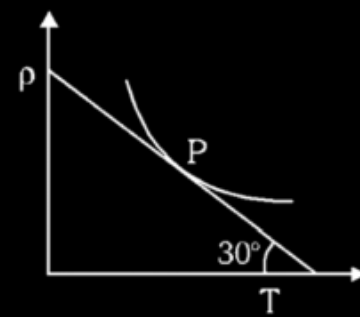


Consider the given circuit. Initially all capacitor were uncharged.



If $V_B - V_A = 1$ volt in steady state, the potential difference across the terminals of battery is

The variation of density (ρ) with temperature (T) of an ideal gas of molecular mass (M) at constant pressure is plotted as shown in figure. The pressure of the gas at state P is $\frac{RT^2}{\sqrt{KM}}$. Find the value of K.



In a vernier calliper 10 vernier scale division coincides with 9 main scale division. Where 1 main scale division is 0.1 cm.

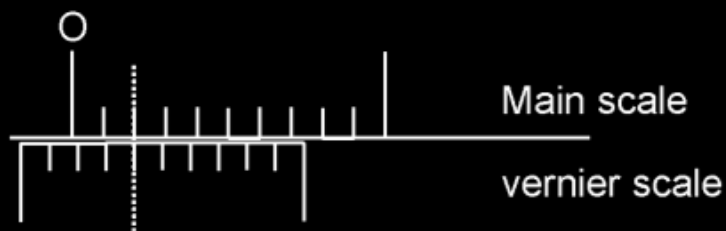


Figure shows main scale and vernier scale when gap between two jaws is zero. The zero error in the faulty vernier calliper is x (in mm) then, $5x$ is